Dear Editor,

We submit our manuscript, "Validation–Deployment Alignment Shapes How Internal Evaluation Predicts External Generalization in Lithium Solid-Electrolyte Machine Learning," for consideration as an Article in the Journal of Chemical Information and Modeling.

Methodological question. Chemistry-family-aware cross-validation is widely recommended as a more realistic internal proxy for out-of-distribution generalization than random k-fold CV, but this premise is rarely tested against a separately curated external dataset for the same property. Prior work constructs out-of-distribution splits within a single dataset, showing that stricter splits are harder, not that they better anticipate external performance. We provide such a test using two separately curated experimental lithium solid-electrolyte conductivity databases (OBELiX and Hargreaves et al.) as a source/target testbed; the two partly share source literature, so this is a lineage-related curation-shift test rather than a fully source-independent pairing. Eight composition-only models, developed on a 449-row OBELiX development population under three internal validation regimes (random five-fold CV, Family-GroupKFold and leave-one-major-family-out), were transferred under an internally pre-specified, hash-locked protocol to a composition-disjoint, 204-row external cohort curated from Hargreaves et al.

What we find. Random CV led all four predeclared regime-comparison statistics on the full external cohort by point estimate; the family-aware regimes did not anticipate external error or model ranking better than random CV, so the pre-specified hypotheses were not supported. Most between-regime confidence intervals include zero, and we report the result as directional rather than decisive. A deployment-scenario stratification qualifies it: for compositions within chemistry families already represented in development data (68% of the external cohort), random CV closely anticipated external error; for an unseen glass/glass-ceramic material class, family-aware CV better anticipated the magnitude of the much larger external error, although it still did not identify the best model. The analogous question for unseen crystalline chemistry remains unresolved at this sample size. Post hoc controls (a training-size-matched randomization, formula- and chemical-system-grouped duplicate controls, a common-support sensitivity, a source/DOI-cluster bootstrap and a family-label sensitivity) show that these conclusions are not artifacts of exact-formula leakage, training-set-size confounding or unequal evaluation populations across regimes.

Significance to JCIM readers. We frame this primarily as a validation-methodology study, not a new conductivity-prediction model. The transferable finding is that internal validation should be matched to the kind of distribution shift expected at deployment, rather than chosen because a split is more restrictive. This is relevant to readers working on model validation, benchmark design and deployment-risk assessment in cheminformatics and materials informatics, and it complements prior work showing that stricter splits give more honest internal accuracy estimates: being more honest internally and being more predictive of external performance are separable properties of a validation regime.

Rigor and reproducibility. The primary external-transfer protocol was frozen and SHA-256 hash-locked before external targets were read. Subsequent post hoc analyses were separately specified and hash-locked before their respective computations. Two implementation corrections made before submission are disclosed in full: a nested-CV correction of the internal validation, and a feature-lookup bug fix for one of the eight models, which received one corrective external refit under the same frozen protocol. Neither changes the conclusions. Code, frozen predictions, evaluation records and a reproducibility manifest accompany the submission. The source databases are used under their respective terms, and the raw Hargreaves database is not redistributed.

Administrative. This work has not been published elsewhere and is not under consideration at another journal. The authors declare no competing interests. We look forward to your consideration.

Sincerely,

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